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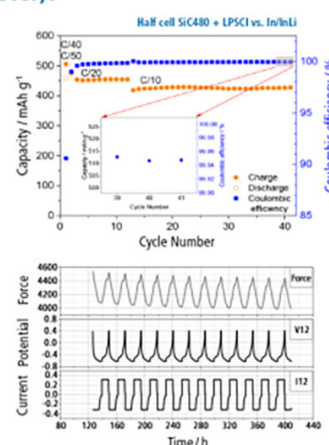
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Dual-Continuum Models of Lithium-Ion Batteries are Fast and Accurate Alternatives to the Doyle-Fuller-Newman Approach: II. Model Extensions

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In Part I of this work, we derived and validated a dual-continuum model of a lithium-ion battery using the volume-averaging technique. Such models employ a fully macroscale description of the battery, thus avoiding the strong assumption made in the Doyle-Fuller-Newman (DFN) model that active material particles are isolated spheres. In all cases studied, our dual-continuum model predicted cell voltage more accurately than the DFN model, and required 70–80% less computation time. The physical insight offered by volume averaging gives rise to several interesting extensions of our derivation. Here in Part II, four of these are considered. Firstly, while Part I relied on a *quasi-steady-state* assumption for the closure problem, we now consider its transient behaviour to improve accuracy under changing battery loads. Secondly, we simulate electrodes with carbon additives and polymer binder, which can be explicitly modelled in the closure problem. Thirdly, we extend the dual-continuum theory to a multi-continuum model, which is particularly interesting for electrodes with a large particle size distribution. Finally, we reduce the dual-continuum theory to two single-continuum formulations, which are comparable with the well-known single-particle model—these maintain much of the accuracy of the dual-continuum approach whilst further reducing computation time.

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The ubiquitous Doyle-Fuller-Newman (DFN) model for lithium-ion batteries can be derived using upscaling theories, such as multiscale asymptotics^{1–3} or the volume-averaging technique (VAT).^{4,5} From the perspective of such theories, it is considered a *hybrid* model,⁶ since the majority of the transport equations are upscaled, but a microscale description of the active material (AM) particles is maintained. Another family of models which rarely appears in the battery-modelling literature is the “dual-continuum” (DC) approach,^{6–8} where *all* equations are upscaled. In such models, the primary challenge is obtaining an estimate for AM surface concentration, since the upscaled equation describes volume-averaged concentration.

In Part I of this work, we developed a DC model of a lithium-ion battery with the VAT, using a closure problem on the electrode microstructure to estimate surface concentration. The model was subsequently validated by comparison with a detailed microscale model, and also with experimental data. Included in this comparison was the DFN model, and also another form of DC model recently published by Yang and Tartakovsky.⁷ In all cases studied, we found that the DC models predicted cell voltage more accurately than the DFN model, and that they required roughly 70–80% less computation time. Nonetheless, Part I showcases a DC model in a relatively simple form, and the technique of volume averaging with closure readily allows for several interesting extensions—four of these are presented here in Part II. Together, these form a progression: complexity, to generality, to efficiency. We increase the complexity of the closure problem, generalise the closure problem to include more complex microstructures, generalise the DC model structure, and increase efficiency using a model reduction.

Transient closure. In Part I, a mapping variable, s_i , was used to relate volume-averaged concentration to surface concentration, facilitating an improved estimate of the reaction rate. A closure problem was solved on the electrode microstructure to determine s_i . The “quasi-steady-state” assumption was employed in the

development of the closure problem, which leads to the mapping variable being independent of time. Instead, this assumption could be avoided, and one could consider the transient behaviour of s_i . Although such models are standard in the upscaling literature,^{6,9} they are often relatively impractical, as they introduce a time convolution to the macroscale equations. However, we demonstrate that in battery modelling a convenient simplification often exists for the time convolution.

Explicit treatment of carbon-binder domain (CBD). Carbon additives and polymer binders are routinely added to electrodes to enhance electronic conductivity and mechanical integrity (respectively), and their interaction often leads to the formation of carbon-binder agglomerates.¹⁰ Both the DFN and Yang’s model have limited means to explicitly model the effect of carbon-binder—only the calculation of effective transport properties and specific surface area will be affected by its presence. By contrast, the DC framework can directly incorporate CBD effects through the closure problem, enabling straightforward modelling of CBD deposits on the AM surface.

Generalisation to multi-continuum models. DC models belong to the wider family of multi-continuum (MC) models (often referred to as “multi-rate mass transfer”¹¹). In this more general approach, one is not limited to describing the AM as a single phase/continuum, and it can instead be decomposed into several phases—this allows the characteristic diffusion times for particles of different sizes to be captured better. In this section, we revisit the derivation in Part I and extend the DC approach to an MC model. Specifically, we average over particle groups rather than the entire particle population, resulting in a model with N macroscale equations describing AM concentration, where N is the number of particle groups.

Reduction to single-continuum models. Throughout both parts of this work, the DC models are benchmarked against the DFN. Another popular choice for modelling batteries is the single-particle model (SPM) family—this includes various different formulations,^{12–14} which are summarised well by Brosa-Planella et al.¹⁵ The primary interest of an SPM is its reduced computational cost, offering a lightweight yet still physics-based alternative to the DFN. They are typically derived by

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further reducing the DFN, using an asymptotic expansion for example.^{12,13} As the name suggests, this results in a model where a single particle represents each electrode (rather than a continuum of particles, as in the DFN). A similar rationale can also be applied to the DC model—rather than describing AM concentration along the thickness of the electrode, an averaged value can be used, gaining a substantial improvement in computational efficiency. The resulting model is therefore “single continuum” rather than dual continuum.

The Transient Dual-Continuum Model (DC^{1t})

Closure.—We re-enter at the section titled “The Closure Problem” in Part I—all derivations prior to this remain identical. In the process of upscaling, the AM concentration (c_{k_i}) for Particle i within the microscale model was decomposed into an average plus a deviation,

$$c_{k_i} = \langle c_{k_i} \rangle^k + \tilde{c}_{k_i}. \quad [1]$$

A partial-differential equation (PDE) was derived for $\tilde{c}_{k_i}$ in order to understand the macroscale source terms causing deviations on the microscale. However, in the original development, the “quasi-steady-state” assumption was made, $\partial_t \tilde{c}_{k_i} \approx 0$. To develop a DC model with the transient behaviour of $\tilde{c}_{k_i}$ included, herein referred to as the transient DC model or DC^{1t}, we begin from the system of equations on $\tilde{c}_{k_i}$ derived in Part I,

$$\partial_t \tilde{c}_{k_i} = \nabla \cdot (D_k \nabla \tilde{c}_{k_i}) + \langle j_{k_i} \rangle_A \frac{1}{F} \frac{A_i}{V_{k_i}} \quad \text{in } \Omega_{k_i}, \quad [2a]$$

$$-\mathbf{n} \cdot D_k \nabla \tilde{c}_{k_i} = \frac{\langle j_{k_i} \rangle_A}{F} \quad \text{at } \partial\Omega_{k_i,e}, \quad [2b]$$

$$\mathbf{n} \cdot D_k \nabla \tilde{c}_{k_i} = \mathbf{n} \cdot D_k \nabla \tilde{c}_{k_j} \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [2c]$$

$$\tilde{c}_{k_i} = \tilde{c}_{k_j} \quad \text{at } \partial\Omega_{k_i,k_j}. \quad [2d]$$

Since the $\partial_t \tilde{c}_{k_i}$ term is now maintained, we cannot proceed with the closure as in Part I. Instead, a Laplace transform is applied to the PDE,

$$p\tilde{c}_{k_i}^* - \tilde{c}_{k_i}|_{t=0} = \nabla \cdot (D_k \nabla \tilde{c}_{k_i}^*) + \frac{A_i}{FV_{k_i}} \langle j_{k_i} \rangle_A^*, \quad [3]$$

where p is the complex frequency variable, and the superscript asterisk denotes the Laplace transform of a function,

$$u^*(p) = \int_0^\infty e^{-pt} u(t) dt. \quad [4]$$

An important assumption is introduced at this point—that $\tilde{c}_{k_i}|_{t=0} = 0$. Physically, this means that the system is initially at rest, and if concentration gradients were to exist at $t = 0$ s then this assumption would not be valid. Using this in Eq. 3, and applying the Laplace transform to the boundary conditions (BCs) gives

$$p\tilde{c}_{k_i}^* = \nabla \cdot (D_k \nabla \tilde{c}_{k_i}^*) + \frac{A_i}{FV_{k_i}} \langle j_{k_i} \rangle_A^* \quad \text{in } \Omega_{k_i}, \quad [5a]$$

$$-\mathbf{n} \cdot D_k \nabla \tilde{c}_{k_i}^* = \frac{\langle j_{k_i} \rangle_A^*}{F} \quad \text{at } \partial\Omega_{k_i,e}, \quad [5b]$$

$$\mathbf{n} \cdot D_k \nabla \tilde{c}_{k_i}^* = \mathbf{n} \cdot D_k \nabla \tilde{c}_{k_j}^* \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [5c]$$

$$\tilde{c}_{k_i}^* = \tilde{c}_{k_j}^* \quad \text{at } \partial\Omega_{k_i,k_j}. \quad [5d]$$

The macroscale source terms present are $\langle j_{k_i} \rangle_A^*$ and $\langle j_{k_j} \rangle_A^*$ (indirectly through the presence of $\tilde{c}_{k_j}^*$). Identically to Part I, we simplify this immediately by arguing that the surface-averaged reaction rate for two neighbouring particles is relatively similar, $\langle j_{k_i} \rangle_A^* \approx \langle j_{k_j} \rangle_A^*$, which motivates us to propose the following closure,

$$\tilde{c}_{k_i}^* = p s_i^{t,*} \langle j_{k_i} \rangle_A^*, \quad [6a]$$

$$\tilde{c}_{k_j}^* = p s_j^{t,*} \langle j_{k_j} \rangle_A^*. \quad [6b]$$

Inserting Eq. 6 into Eq. 5 gives the following closure problem,

$$p s_i^{t,*} = \nabla \cdot (D_k \nabla s_i^{t,*}) + \frac{1}{p} \frac{A_i}{FV_{k_i}} \quad \text{in } \Omega_{k_i}, \quad [7a]$$

$$-\mathbf{n} \cdot D_k \nabla s_i^{t,*} = \frac{1}{pF} \quad \text{at } \partial\Omega_{k_i,e}, \quad [7b]$$

$$\mathbf{n} \cdot D_k \nabla s_i^{t,*} = \mathbf{n} \cdot D_k \nabla s_j^{t,*} \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [7c]$$

$$s_i^{t,*} = s_j^{t,*} \quad \text{at } \partial\Omega_{k_i,k_j}. \quad [7d]$$

After transforming back into the time domain, the final closure problem is,

$$\partial_t s_i^t = D_k \Delta s_i^t + \frac{1}{F} \frac{A_i}{V_{k_i}} \quad \text{in } \Omega_{k_i}, \quad [8a]$$

$$-\mathbf{n} \cdot D_k \nabla s_i^t = \frac{1}{F} \quad \text{at } \partial\Omega_{k_i,e}, \quad [8b]$$

$$\mathbf{n} \cdot \nabla s_i^t = \mathbf{n} \cdot \nabla s_j^t \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [8c]$$

$$s_i^t = s_j^t \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [8d]$$

$$s_i^t = 0 \quad \text{at } t = 0. \quad [8e]$$

Finally, transforming the expression for $\tilde{c}_{k_i}^*$ (Eq. 6a) into the time domain gives

$$\tilde{c}_{k_i} = \partial_t s_i^t \star \langle j_{k_i} \rangle_A, \quad [9]$$

where the time convolution $f \star g$ is defined as

$$f \star g = \int_0^t f(\tau) g(t - \tau) d\tau. \quad [10]$$

Physically, this means that the solution for $\tilde{c}_{k_i}$ depends on the value of $\langle j_{k_i} \rangle_A$ not only at time t but also at all previous times.^{6,16}

Macroscale.—Here, we derive the macroscale PDE which will describe the evolution of AM concentration. Using the decomposition of c_{k_i} (Eq. 1) and the closure developed above (Eq. 9), the microscale AM concentration field can be reconstructed,

$$c_{k_i} = \langle c_{k_i} \rangle^k + \partial_t s_i^t \star \langle j_{k_i} \rangle_A, \quad [11]$$

which can then be exploited to build an approximation of surface concentration,

$$\langle c_k \rangle_A \equiv \frac{1}{A} \sum_{i=1}^n \int_{A_i} c_{k,i} dA. \quad [12]$$

With identical assumptions to the “First-Order Macroscale Model” Section in Part I—notably that $\langle j_{k,i} \rangle_A$ does not vary significantly across the population of particles within an averaging volume, $\langle j_{k,i} \rangle_A \approx \langle j_k \rangle_A$ —we arrive at the following surface concentration for DC^{1r},

$$\langle c_k \rangle_A^{1r} = \langle c_k \rangle^k + \langle s^t \rangle_A \star \partial_t \langle j_k \rangle_A, \quad [13]$$

where,

$$\langle s^t \rangle_A = \frac{1}{A} \sum_{i=1}^n \int_{A_i} s_i^t dA. \quad [14]$$

Note that the analytic properties of convolutions have been exploited above to interchange the time derivative. We now use this estimate of surface concentration to define an improved estimate of reaction rate, j_k^{1r} ,

$$j_k^{1r} = 2j_{k0}^{1r} \sinh\left(\frac{F}{2RT} \eta_k^{1r}\right), \quad [15]$$

where,

$$\eta_k^{1r} = \langle \phi_k \rangle^k - \langle \phi_e \rangle^e - U_k(\langle c_k \rangle_A^{1r}), \quad [16a]$$

$$j_{k0}^{1r} = Fk_{am} \sqrt{\langle c_k \rangle_A^{1r} \langle c_e \rangle^e (c_{k,\max} - \langle c_k \rangle_A^{1r})}. \quad [16b]$$

Since we do not know $\langle j_k \rangle_A$ a priori, we assume that $\langle j_k \rangle_A \approx j_k^{1r}$,

$$\langle c_k \rangle_A^{1r} = \langle c_k \rangle^k + \langle s^t \rangle_A \star \partial_t j_k^{1r}. \quad [17]$$

The macroscale PDE thus takes the following form,

$$\partial_t \langle c_k \rangle^k = -\frac{A}{FV_k} j_k^{1r}. \quad [18]$$

Although this model is solvable, a convolution appears in our macroscale equation (via j_k^{1r}), which significantly complicates the model. However, an approximation for the convolution may be developed which is applicable for galvanostatic cycling profiles. Consider the scenario of a constant-current (dis)charge beginning at time 0 s followed by a relaxation beginning at time t_r . Here, $\langle j_k \rangle_A$ is invariant with space and time until the point of the relaxation, where it then rapidly decays to 0 A m⁻². Thus, the function $\langle j_k \rangle_A$ can be represented by a Heaviside step function, $H(t)$, making its derivative the Dirac delta function, $\delta(t)$,

$$\langle j_k \rangle_A(t) = \langle j_k \rangle_A^c [H(t) - H(t - t_r)], \quad [19]$$

where $\langle j_k \rangle_A^c$ is the constant value of the averaged reaction current density during the (dis)charge, $0 < t < t_r$, given by

$$\langle j_k \rangle_A^c = \lambda_k \frac{i_{\text{app}}}{L_k a_k}, \quad [20]$$

where L_k is the length of the electrode, a_k is the specific surface area, $\lambda_n = -1$ and $\lambda_p = 1$. This is identical to the treatment of the reaction rate in the Single-Particle Model developed by Marquis et al.¹² The time derivative of $\langle j_k \rangle_A$ is now,

$$\partial_t \langle j_k \rangle_A(t) = \langle j_k \rangle_A^c [\delta(t) - \delta(t - t_r)], \quad [21]$$

and Eq. 13 becomes

$$\langle c_k \rangle_A^{1r} = \langle c_k \rangle^k + \langle s^t \rangle_A \star \langle j_k \rangle_A^c [\delta(t) - \delta(t - t_r)]. \quad [22]$$

Exploiting the algebraic properties of convolutions (including the fact that convolving a function with δ gives the original function itself),

$$\langle c_k \rangle_A^{1r} = \langle c_k \rangle^k + \langle j_k \rangle_A^c [\langle s^t \rangle_A(t) - \langle s^t \rangle_A(t - t_r)]. \quad [23]$$

We may write this as,

$$\langle c_k \rangle_A^{1r} = \begin{cases} \langle c_k \rangle^k + \langle j_k \rangle_A^c \langle s^t \rangle_A(t) & \text{for } 0 < t < t_r, \\ \langle c_k \rangle^k + \langle j_k \rangle_A^c [\langle s^t \rangle_A(t) - \langle s^t \rangle_A(t - t_r)] & \text{for } t_r \leq t. \end{cases} \quad [24]$$

This can easily be extended for an arbitrary number of constant-current operations,

$$\langle j_k \rangle_A^{c,i} = \lambda_k \frac{i_{\text{app},i}}{L_k a_k}, \quad [25a]$$

$$\langle j_k \rangle_A(t) = \sum_{i=1}^n \langle j_k \rangle_A^{c,i} [H(t - t_{s,i}) - H(t - t_{e,i})], \quad [25b]$$

$$\langle c_k \rangle_A^{1r} = \langle c_k \rangle^k + \sum_{i=1}^n \langle j_k \rangle_A^{c,i} [\langle s^t \rangle_A(t - t_{s,i}) - \langle s^t \rangle_A(t - t_{e,i})], \quad [25c]$$

where $i_{\text{app},i}$ is the constant current density applied at the current collector for operation i , whilst $t_{s,i}$ and $t_{e,i}$ are the start and end times of the operation (respectively). Note that for constant-voltage operations, this simplification is not valid and the convolution should be maintained (Eq. 17).

Validation methodology.—The model with the transient closure, DC^{1r}, was parameterised and solved for the 3D nickel manganese cobalt oxide (NMC) half cell presented in Part I of this work (see Fig. 4 in Part I). Other than the closure variable $\langle s^t \rangle_A$, all parameters remain identical to those used in Part I (material properties and solver choices are unchanged). Likewise, as in Part I, OpenFOAM’s chtMultiRegionFoam solver was used to solve the closure problem (Eq. 8).

Two cycling profiles were simulated for this half cell: a 1C discharge to 3 V followed by a 15 minute relaxation (Profile 1), and a series of discharge/relaxation operations (Profile 2). For Profile 2, each discharge operation was terminated when a voltage constraint was reached (3.8 V, 3.7 V, 3.6 V, 3.5 V, and 3.4 V). Between each discharge, the cell was left to relax for 50 s, whilst the final relaxation period was 200 s. The macroscale models solved include the DFN, DC^y, DC¹, and DC^{1r}. All macroscale models are evaluated against the direct numerical solution (DNS) of the microscale model presented in Part I.

As a final validation step, we repeated the simulations shown in Fig. 6b in Part I, in which a lower diffusivity ($D_k = 8 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$) was used and the discharge was terminated at 3.5 V—DC^{1r} is now included in the comparison.

Results and discussion.—The behaviour of $\langle s^t \rangle_A$ with time is shown in Fig. 1a, whilst the cell-voltage for Profiles 1 and 2 are shown in Figs. 1b, 2 and 3a. Beginning with Profile 1, DC^{1r} is able to

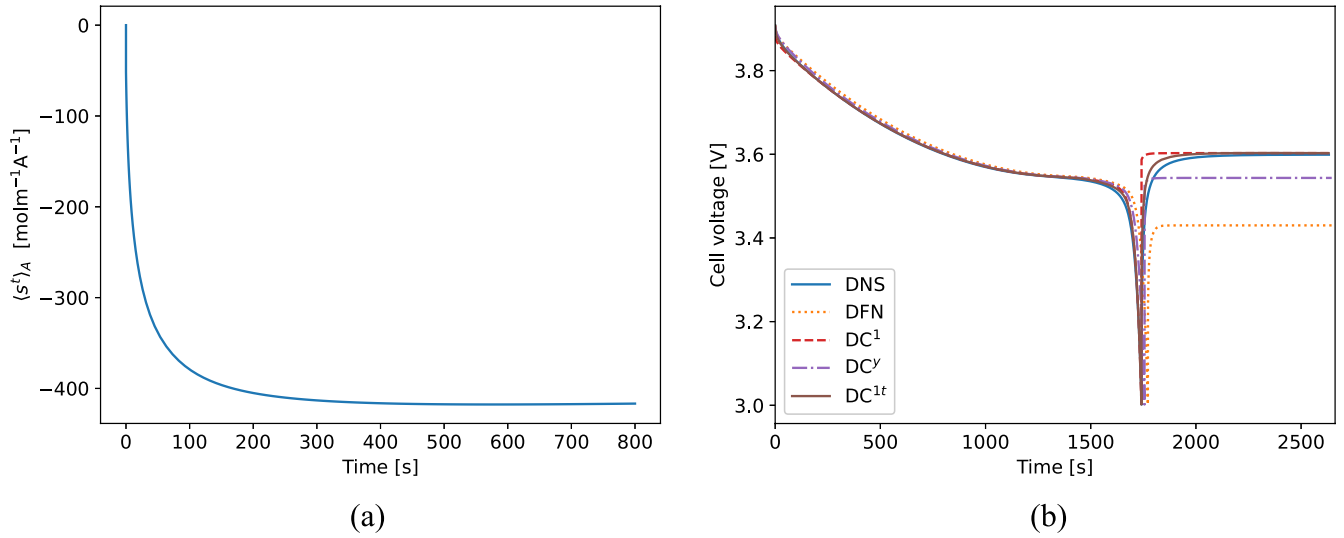


Figure 1. The relationship of $\langle s^i \rangle_A$ with time obtained when Eq. 8 is solved on Electrode 3D (a), and cell-voltage against time for a 1C discharge-relaxation simulation on Electrode 3D, showing the DNS, DFN, DC¹, DC^y, and DC^{1t} (b).

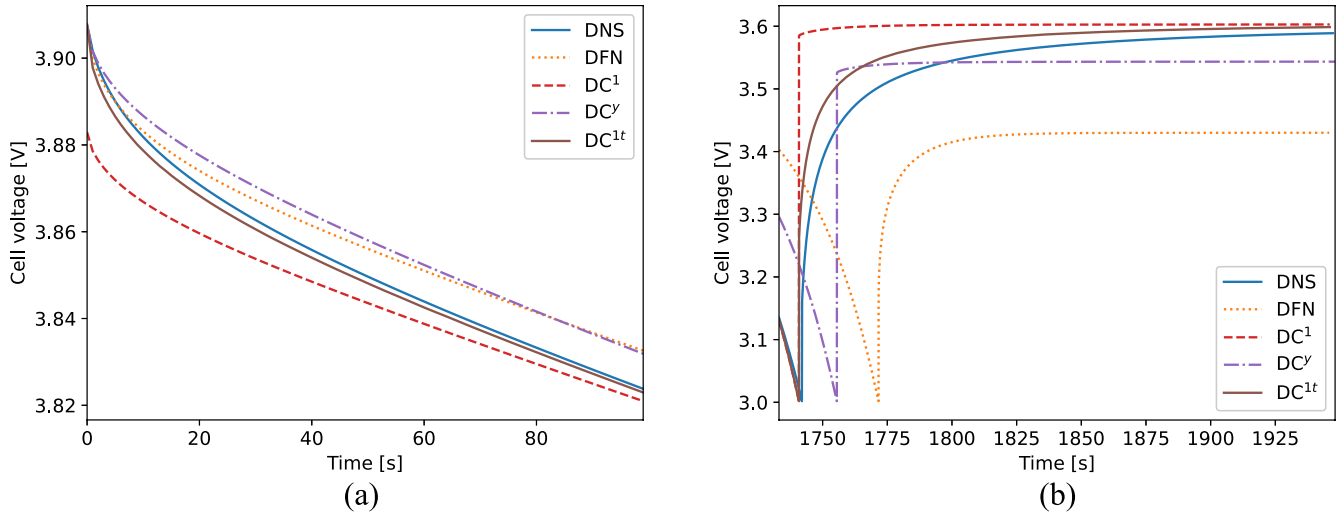


Figure 2. Enlarged views of the beginning (a) and the end (b) of the 1C discharge shown in Fig. 1b.

capture the start of the discharge better than DC¹—this is more clear in Fig. 2a, where the interval of 0 s–100 s is enlarged. This improvement is thanks to the gradual increase in $\langle c_k \rangle_A^{1t}$ realised by DC^{1t}. Figure 1a shows the value of $\langle s^i \rangle_A$ is small at early times and grows toward a steady-state value. Therefore, in DC^{1t} the difference between surface and bulk concentration, $\langle c_k \rangle_A^{1t} - \langle c_k \rangle^k$, also grows gradually to a steady-state value (see Eq. 24). On the other hand, in DC¹ only the steady-state value of $\langle s \rangle_A$ is used, and thus the surface concentration, $\langle c_k \rangle_A^1$, is overestimated at early times.

For a similar reason, DC^{1t} is also able to better capture the relaxation behaviour. Here, it models the gradual decrease of $\langle c_k \rangle_A^{1t}$ to $\langle c_k \rangle^k$ once the cell is left to relax. On the other hand, $\langle c_k \rangle_A^1$ rapidly decays to $\langle c_k \rangle^k$ in DC¹ (since $\langle j_k \rangle_A$ rapidly decays to 0 A m⁻²), and hence it does not capture the slow relaxation visible in the DNS.

Nonetheless, these results show that the quasi-steady-state assumption employed in DC¹ is fairly reasonable—DC¹ and DC^{1t} overlap completely for the majority of the simulation. This is thanks to the time taken for s_i to approach its steady-state value being short compared to the battery discharge time.

When AM diffusivity is reduced, the advantage of DC^{1t} over DC¹ is emphasised. In Fig. 3b, DC¹ exhibits a considerable error at the

start of the discharge (as pointed out in Part I). Nonetheless, the inclusion of the transient closure almost entirely removes this error. This is justified by the fact that s_i takes longer to reach its steady-state value when D_k is lower, and thus ignoring its variation with time is less reasonable.

To some degree, the DFN is able to capture the cell-voltage behaviour at the beginning of the discharge and during the relaxation. Since the DFN solves for c_k on an auxiliary radial dimension at each point along the electrode thickness, it is able to naturally model the slow increase/relaxation of surface concentration. The errors in the DFN results (compared to the DNS) are instead primarily due to its assumptions on particle shape, as discussed in Part I.

Likewise, DC^y does include means to capture the slow increase of surface concentration at the beginning of the discharge, since the thickness of the boundary layer in the model is a function of time. Nonetheless, Fig. 2a reveals that DC^{1t} is considerably more accurate than DC^y in this period, which is due to the deficiencies of boundary layer theory mentioned in Part I. Additionally, DC^y does not model the slow decrease in surface concentration during the relaxation period, hence it is not able to capture the slow voltage relaxation seen in the DNS and DC^{1t} (Fig. 2b).

In terms of practicality, the addition of the transient closure does not complicate the DC model significantly, and the user only needs

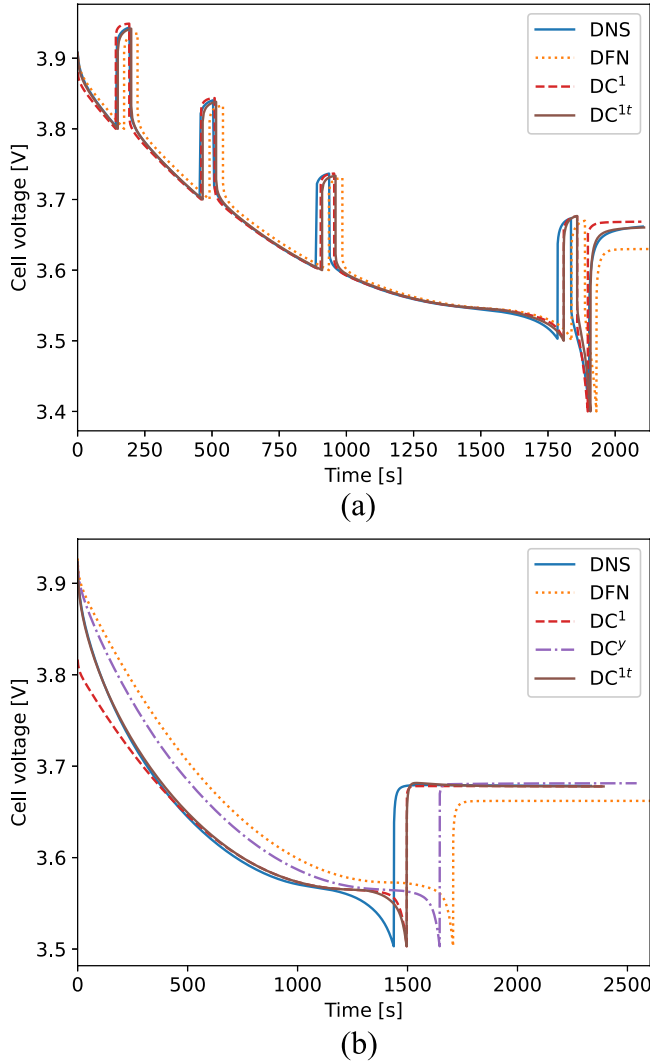


Figure 3. Cell-voltage against time for Profile 2 on Electrode 3D (a) and for Electrode B with a lower AM diffusivity (b), $D_k = 8 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$.

to tabulate the value of $\langle s^t \rangle_A(t)$ to parameterise DC^{1t} . It also does not increase the computation time, and for this simulation DC^1 and DC^{1t} took 0.0733s and 0.0717s respectively (see Part I supplementary material for calculation details).

Profile 2 was included to demonstrate the theory of DC^{1t} being extended to an arbitrary number of constant-current operations (Eq. 25). Figure 3a shows that DC^{1t} remains accurate for the later operations, and suffers from significantly less voltage “drift” compared to the DFN, as it predicts the end times of each operation more precisely. DC^y has not been included in this plot to avoid clutter.

Accounting for CBD

Theory.—The microscale model from Part I of this work can be extended by including a carbon-binder domain (CBD), Ω_{cbd} , which is an effective phase containing both carbon/binder agglomerates and electrolyte.^{17–20} Therefore, the CBD enables electrolyte mass flux ($N_{e,\text{cbd}}$), electrolyte current ($i_{e,\text{cbd}}$), and electronic current ($i_{k,\text{cbd}}$). We define the AM–CBD interface as $\partial\Omega_{k,\text{cbd}}$, the electrolyte–CBD interface as $\partial\Omega_{e,\text{cbd}}$, whilst $\partial\Omega_k = \partial\Omega_{k,e} \cup \partial\Omega_{k,\text{cbd}}$. This phase is considered to have both a volume porosity ($\varepsilon_{e,\text{cbd}}$), and also a surface porosity ($\varepsilon_{e,\text{cbd}}^{\text{surf}}$), where the latter is the fraction of $\partial\Omega_{k,\text{cbd}}$ available for reaction (as in Hein’s work¹⁷). Likewise, the electrolyte phase within the CBD is tortuous, and thus has a tortuosity, $\tau_{e,\text{cbd}}$, and a

transport efficiency, $\mathfrak{B}_{e,\text{cbd}} = \varepsilon_{e,\text{cbd}} \tau_{e,\text{cbd}}^{-1}$. The mass conservation in the CBD is given by,

$$\varepsilon_{e,\text{cbd}} \partial_t c_{e,\text{cbd}} + \nabla \cdot N_{e,\text{cbd}} = 0 \quad \text{in } \Omega_{\text{cbd}}, \quad [26a]$$

$$\mathbf{n} \cdot N_{e,\text{cbd}} = \varepsilon_{e,\text{cbd}}^{\text{surf}} F^{-1} j_k \quad \text{at } \partial\Omega_{k,\text{cbd}}, \quad [26b]$$

$$\mathbf{n} \cdot N_k = \varepsilon_{e,\text{cbd}}^{\text{surf}} F^{-1} j_k \quad \text{at } \partial\Omega_{k,\text{cbd}}, \quad [26c]$$

$$\mathbf{n} \cdot N_{e,\text{cbd}} = \mathbf{n} \cdot N_e \quad \text{at } \partial\Omega_{e,\text{cbd}}, \quad [26d]$$

$$c_{e,\text{cbd}} = c_e \quad \text{at } \partial\Omega_{e,\text{cbd}}, \quad [26e]$$

whilst charge conservation yields,

$$\nabla \cdot i_{e,\text{cbd}} = 0 \quad \text{in } \Omega_{\text{cbd}}, \quad [27a]$$

$$\nabla \cdot i_{k,\text{cbd}} = 0 \quad \text{in } \Omega_{\text{cbd}}, \quad [27b]$$

$$\mathbf{n} \cdot i_{e,\text{cbd}} = \varepsilon_{e,\text{cbd}}^{\text{surf}} j_k \quad \text{at } \partial\Omega_{k,\text{cbd}}, \quad [27c]$$

$$\mathbf{n} \cdot i_k = \mathbf{n} \cdot (i_{k,\text{cbd}} + i_{e,\text{cbd}}) \quad \text{at } \partial\Omega_{k,\text{cbd}}, \quad [27d]$$

$$\mathbf{n} \cdot i_{e,\text{cbd}} = \mathbf{n} \cdot i_e \quad \text{at } \partial\Omega_{e,\text{cbd}}, \quad [27e]$$

$$\mathbf{n} \cdot i_{k,\text{cbd}} = 0 \quad \text{at } \partial\Omega_{e,\text{cbd}}, \quad [27f]$$

$$\phi_{e,\text{cbd}} = \phi_e \quad \text{at } \partial\Omega_{e,\text{cbd}}, \quad [27g]$$

$$\phi_{k,\text{cbd}} = \phi_k \quad \text{at } \partial\Omega_{k,\text{cbd}}. \quad [27h]$$

The following expressions are used to describe mass and charge flux,

$$N_{e,\text{cbd}} = -\mathfrak{B}_{e,\text{cbd}} D_e \nabla c_{e,\text{cbd}} + \frac{t_+^0}{F} i_{e,\text{cbd}}, \quad [28a]$$

$$i_{e,\text{cbd}} = -\mathfrak{B}_{e,\text{cbd}} \sigma_e \nabla \phi_{e,\text{cbd}} - \mathfrak{B}_{e,\text{cbd}} \sigma_e \partial \nabla \ln c_{e,\text{cbd}}, \quad [28b]$$

$$i_{k,\text{cbd}} = -\sigma_{k,\text{cbd}} \nabla \phi_{k,\text{cbd}}, \quad [28c]$$

where $c_{e,\text{cbd}}$ and $\phi_{e,\text{cbd}}$ are the concentration and potential (respectively) of electrolyte within the CBD. Finally, $\phi_{k,\text{cbd}}$ and $\sigma_{k,\text{cbd}}$ are the solid-phase potential and electronic conductivity (respectively) of the CBD.

Volume averaging and closure.—To develop a closure problem, we follow an analogous method to Part I—the key difference is that we now have an additional BC, at $\partial\Omega_{k,\text{cbd}}$. With the addition of this BC, the system of equations on c_{k_i} from (Eq. 11 in Part I) becomes,

$$\partial_t c_{k_i} = \nabla \cdot (D_k \nabla c_{k_i}) \quad \text{in } \Omega_{k_i}, \quad [29a]$$

$$-\mathbf{n} \cdot D_k \nabla c_{k_i} = F^{-1} j_{k_i} \quad \text{at } \partial\Omega_{k,e}, \quad [29b]$$

$$-\mathbf{n} \cdot D_k \nabla c_{k_i} = \varepsilon_{e,\text{cbd}}^{\text{surf}} F^{-1} j_{k_i} \quad \text{at } \partial\Omega_{k,\text{cbd}}. \quad [29c]$$

The homogenised equation on $\langle c_{k_i} \rangle^k$ and the PDE on $\tilde{c}_{k_i}$ developed in Part I are still applicable, however the definition of the surface average operator $\langle u \rangle_A$ and A_i are now,

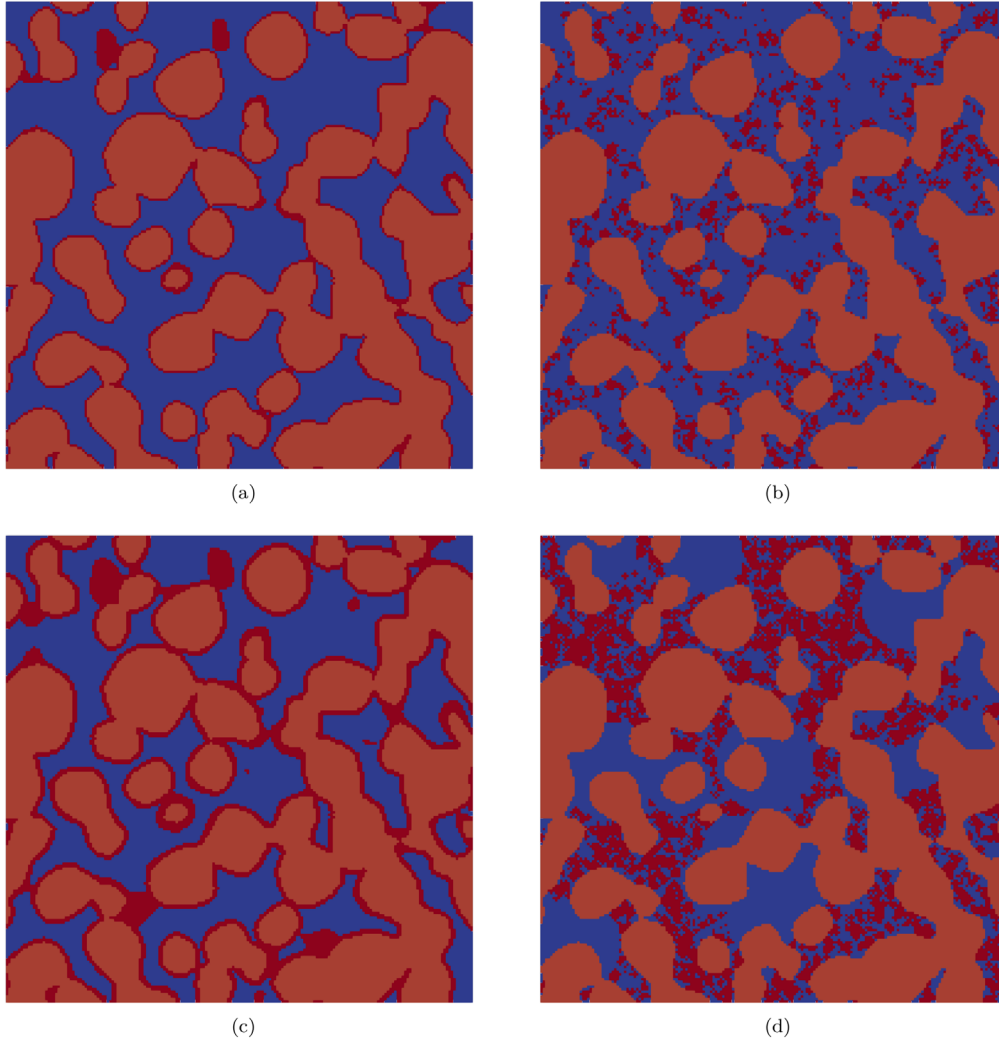


Figure 4. Illustrating the CBD generation algorithms, using an arbitrary 2D slice from the centre of each electrode: a) Surface-10, b) Clusters-10, c) Surface-20, and d) Clusters-20.

$$\langle u \rangle_A \equiv \frac{1}{A_i} \left(\int_{\mathcal{A}_{ei}} u dA + \varepsilon_{e,\text{cbd}}^{\text{surf}} \int_{\mathcal{A}_{\text{cbd}_i}} u dA \right), \quad [30a]$$

$$A_i = A_{ei} + \varepsilon_{e,\text{cbd}}^{\text{surf}} A_{\text{cbd}_i}, \quad [30b]$$

where the internal boundaries in the averaging set are defined as $\mathcal{A}_{ei}(\mathbf{x}) \subset \partial\Omega_{k,e}$, $\mathcal{A}_{\text{cbd}_i}(\mathbf{x}) \subset \partial\Omega_{k,\text{cbd}}$ and $\mathcal{A}_i(\mathbf{x}) = \mathcal{A}_{ei} \cup \mathcal{A}_{\text{cbd}_i}$. A similar closure problem is reached, but with the addition of an extra BC,

$$\Delta s_i = -\frac{1}{FD_k} \frac{A_i}{V_{k_i}} \quad \text{in } \Omega_{k_i}, \quad [31a]$$

$$\mathbf{n} \cdot \nabla s_i = -\frac{1}{FD_k} \quad \text{at } \partial\Omega_{k_i,e}, \quad [31b]$$

$$\mathbf{n} \cdot \nabla s_i = -\varepsilon_{e,\text{cbd}}^{\text{surf}} \frac{1}{FD_k} \quad \text{at } \partial\Omega_{k_i,\text{cbd}}, \quad [31c]$$

$$\mathbf{n} \cdot \nabla s_i = \mathbf{n} \cdot \nabla s_j \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [31d]$$

$$s_i = s_j \quad \text{at } \partial\Omega_{k_i,k_j}. \quad [31e]$$

An example solution to the closure problem is shown in Fig. 5. Finally, the macroscale equation of DC¹ is identical to that developed previously—we have merely updated the method to calculate $\langle s \rangle_A$. For the DFN and all DC models, we adopt the standard approach of assuming local equilibrium between the electrolyte and CBD phases.^{20,21} The two phases are thus homogenised to obtain a single electrolyte transport efficiency, $\mathfrak{B}_e$. Likewise, the charge transport within the AM and CBD is homogenised to obtain a single effective conductivity, σ_k^{eff} .^{19,22}

Validation methodology.—The CBD phase was generated using an in-house algorithm, and two morphologies were tested. In the first, dubbed “Surface”, a film of CBD is created on the AM surface. This type of CBD morphology is very similar to that studied by Usseglio-Viretta,²⁰ Mistry,²³ and Hein¹⁷ (see the electrode also dubbed “Surface” in Hein’s work). In the second, called “Clusters”, CBD clusters are placed stochastically within (most of) the electrolyte phase, but deposition is prohibited in some locations, to recreate the “pore” effect seen in the FIB-SEM images published by Daemi,²⁴ Lu,²⁵ Vierrath,²⁶ and Prifling.¹⁸ 100 pores of radius 7.96

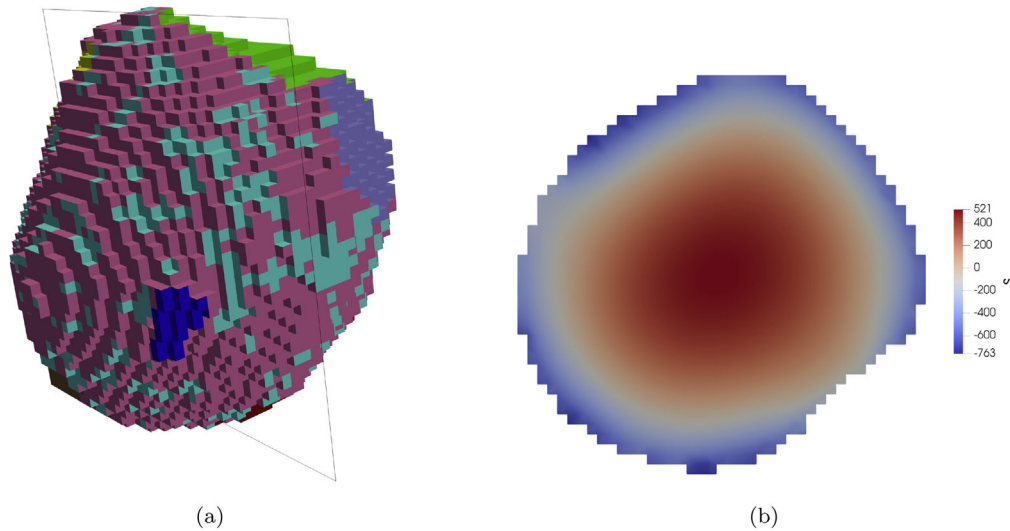


Figure 5. a) An arbitrary particle from Clusters-10: pink is $\partial\Omega_{k_i,e}$, light blue is $\partial\Omega_{k_i,cdb}$, whilst all other colours are $\partial\Omega_{k_i,k_j}$ (for various different j values). b) The steady-state solution to the closure problem on the slice whose outline is shown in Sub-Figure a (s_i units are $\text{mol}\cdot\text{m}^{-1}\cdot\text{A}^{-1}$).

Table I. Effective properties for each electrode and key parameters for DC^I, DC^y, and DFN.

Parameter	Surface-10	Surface-20	Clusters-10	Clusters-20
AM volume fraction			0.493	
Particle radius [μm]			3.405	
γ [$\text{mol}\cdot\text{m}^{-1}\cdot\text{A}^{-1}$]			-294.1	
$\langle s \rangle_A$ [$\text{mol}\cdot\text{m}^{-1}\cdot\text{A}^{-1}$]	-230.1	-230.1	-381.6	-374.4
Effective conductivity, σ_p^{eff} [Sm^{-1}]	0.322	24.3	0.0298	11.0
Electrolyte transport efficiency, $\mathfrak{B}_{e,p}$	0.207	0.133	0.176	0.101
Specific surface area, a_p [m^{-1}]	397026	397026	358046	343465

μm were created. This algorithm produces electrodes similar to Hein's "Contact" morphology. The electrode studied above (Electrode 3D from Part I) was used as an NMC skeleton, and four CBD phases were built—two total CBD volume fractions (0.1 and 0.2) for each morphology (shown in Fig. 4). The electrodes are named such that Surface-10 refers to the electrode with a CBD phase volume fraction of 0.1 (10%) generated using the Surface algorithm.

For the properties of the CBD phase, $\epsilon_{e,cdb}^{\text{surf}} = 0.5$,^{17,18} $\sigma_{k,cdb} = 375 \text{ Sm}^{-1}$,^{27,28} and $\mathfrak{B}_{e,cdb} = 0.0833$ were used (the latter corresponds to $\epsilon_{e,cdb} = 0.5$ ¹⁷ and $\tau_{e,cdb} = 10^{29}$). As in Part I, effective transport properties for the macroscale models were calculated directly from the electrode microstructure, by solving Laplace's equation³⁰ with the in-house code—the method used to include a CBD phase in such simulations has been detailed by Ghadban et al.²¹ Taufactor³¹ was also used to verify results, in which a CBD phase can be explicitly considered by using the 3-phase solver feature. Finally, A_{e_i} and A_{cbd_i} were calculated directly from the electrode images, summing the area of voxel faces at $\partial\Omega_{k,e}$ and $\partial\Omega_{k,cdb}$. We highlight that the capacity of each electrode is identical (since only the CBD phase is being altered), and is calculated using the AM volume from the electrode image (as in Part I). Effective properties and key parameters for each model are summarised in Table I.

To compare the models on these half cells, a 1C discharge to 3V followed by a 15 minute relaxation was simulated. All material properties and numerical settings are identical to those described in Part I, with the exception that the AM conductivity is now set to 0.03 Sm^{-1} , which is a more realistic value for NMC.³² Since CBD was not present in the previous simulations, a value of 2.8 Sm^{-1} was used as an effective solid phase conductivity, to avoid excessive Ohmic losses, as in Hein's work.¹⁷

Results and discussion.—The cell-voltage results for all electrodes are shown in Fig. 6. Although the DFN is reasonably accurate, the DC models possess a clear advantage. For Surface-10 and Surface-20, the DFN shows a voltage offset throughout the entire discharge, which can have a significant impact on performance indicators, particularly cell energy density (where voltage is integrated over time). As in previous simulations, DC^I primarily excels over DC^y toward the end of the discharge and in the relaxation regime. Analogously to the electrode with no CBD, DC^I improves on DC^I at the beginning of discharge and during the relaxation, and is able to almost perfectly reproduce the DNS results for Surface-10 and Surface-20.

The DFN has very little means to account for the effects of CBD: the specific surface area can be reduced to consider surface coverage of carbon/binder, and the effective transport properties can be adjusted to reflect CBD's effect on ion and electron movement, but beyond this the model is limited. Likewise, DC^y (in its current form) does not explicitly consider the CBD, and accounts for its presence through the same effective parameters as the DFN. On the other hand, DC^I introduces a new effective parameter, $\langle s \rangle_A$, which directly models the placement of CBD within the microstructure and its effect on diffusion in the AM. Nonetheless, it is clear that both the DFN and DC^y still produce a good prediction of the DNS results, suggesting that the effective transport properties and specific surface area alone can reasonably capture the CBD's influence.

Multi-Continuum (MC) Models

Theory.—To develop an MC model, we re-enter the derivation of Part I at the "Macroscale" Section—all theory prior to this, and notably the closure problem, remains unchanged. To derive DC^I, we

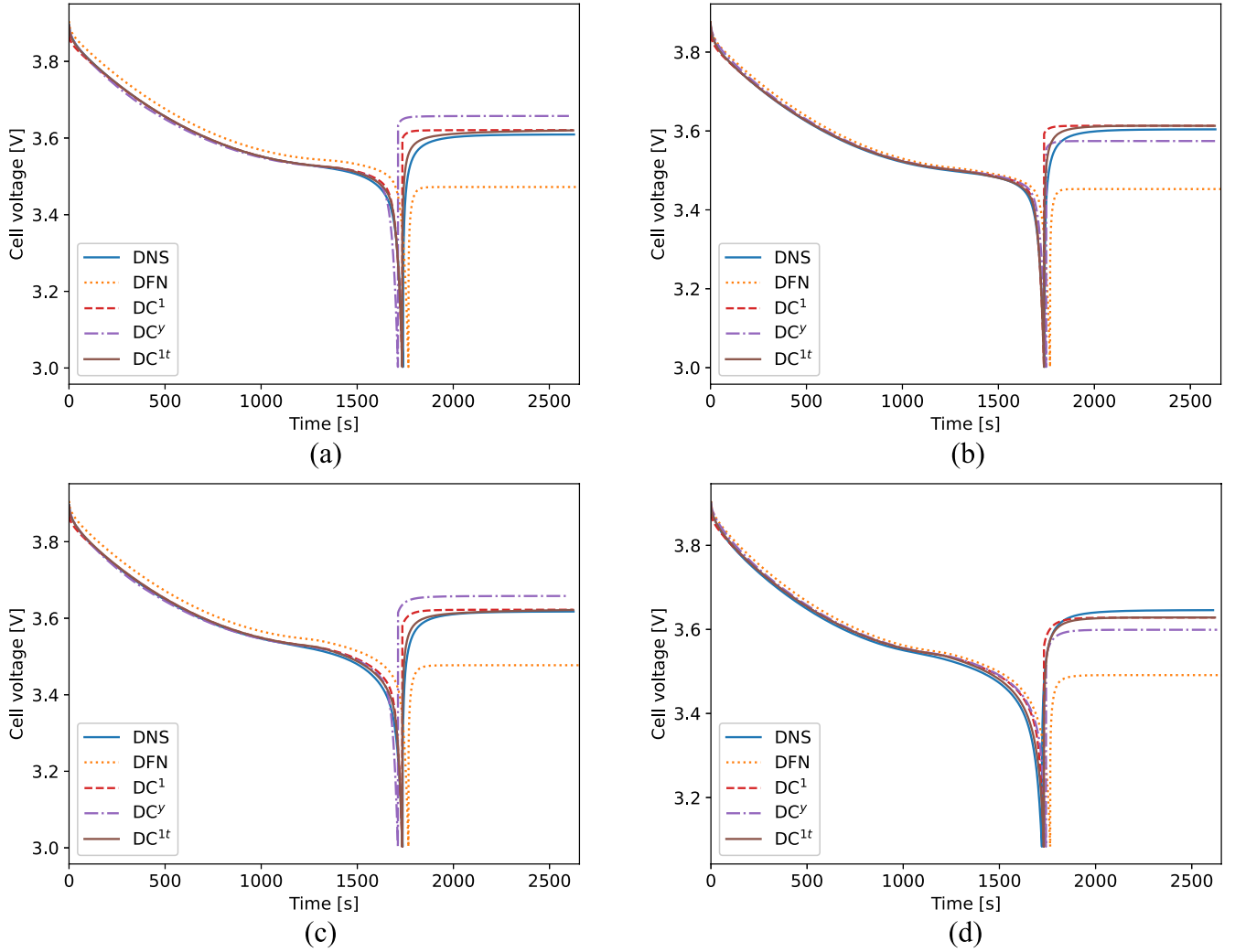


Figure 6. Cell-voltage against time for 1C discharge-relaxation simulations on Surface-10 (a), Clusters-10 (b), Surface-20 (c), and Clusters-20 (d), showing the DNS, DFN, DC^1 , DC^y and DC^{1t} .

assumed that the population of particles in an averaging volume could be represented by the population measured across the entire electrode. Here, this assumption is relaxed. We begin by returning to the volume-averaged mass balance on a per-particle basis, Eq. 22 in Part I,

$$\partial_t \langle c_{k_i} \rangle^k = -\frac{A_i}{FV_{k_i}} \langle j_{k_i} \rangle_A. \quad [32]$$

We now define several particle groups—the aim is to develop a macroscale equation describing each group. We write Eq. 32 as,

$$\frac{V_{k_i}}{V_{k_\alpha}} \partial_t \langle c_{k_i} \rangle^k = -\frac{A_i}{FV_{k_\alpha}} \langle j_{k_i} \rangle_A, \quad [33]$$

where V_{k_α} is the total volume of particles in Group α . Rather than summing across all particles in Ω_k (as in Part I), we will now sum across the particles within each group

$$\sum_{i \in G_\alpha} \frac{V_{k_i}}{V_{k_\alpha}} \partial_t \langle c_{k_i} \rangle^k = -\sum_{i \in G_\alpha} \frac{A_i}{FV_{k_\alpha}} \langle j_{k_i} \rangle_A, \quad [34]$$

where G_α is the set containing the indexes of all particles within Group α . We define

$$\langle c_{k_\alpha} \rangle^k \equiv \sum_{i \in G_\alpha} \frac{V_{k_i}}{V_{k_\alpha}} \langle c_{k_i} \rangle^k, \quad [35a]$$

$$V_{k_\alpha} \equiv \sum_{i \in G_\alpha} V_{k_i}, \quad [35b]$$

$$A_\alpha \equiv \sum_{i \in G_\alpha} A_i, \quad [35c]$$

which can be inserted into Eq. 34 to give

$$\partial_t \langle c_{k_\alpha} \rangle^k = -\frac{1}{FV_{k_\alpha}} \sum_{i \in G_\alpha} A_i \langle j_{k_i} \rangle_A. \quad [36]$$

We now must approximate $\langle j_{k_i} \rangle_A$ to be able to solve the problem. As in Part I, to improve our estimate of reaction rate, we seek to build an approximation of the surface concentration using the mapping variable, which will then be used in the Butler-Volmer expression. Recalling the mapping between $\langle c_{k_i} \rangle^k$ and c_{k_i} ,

$$c_{k_i} = \langle c_{k_i} \rangle^k + s_i \langle j_{k_i} \rangle_A, \quad [37]$$

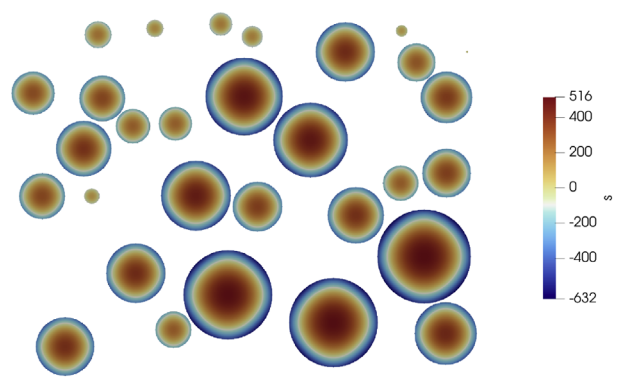
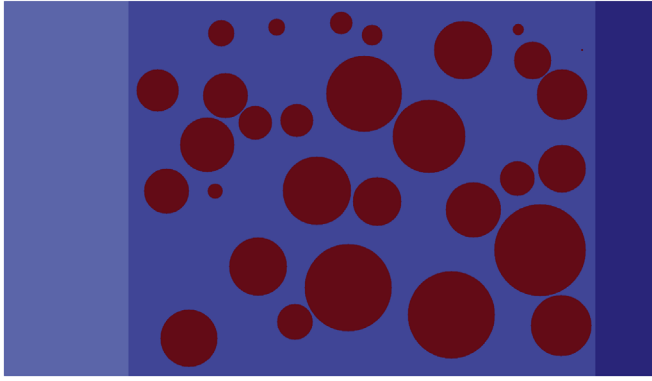


Figure 7. Electrode C (left) and its closure variable solution, s_i (right).

we write this in a different form to illustrate an assumption,

$$c_{k_i} = \langle c_{k_\alpha} \rangle^k + (\langle c_{k_i} \rangle^k - \langle c_{k_\alpha} \rangle^k) + s_i \langle j_{k_i} \rangle_A, \quad [38]$$

and assume that $(\langle c_{k_i} \rangle^k - \langle c_{k_\alpha} \rangle^k) \ll s_i \langle j_{k_i} \rangle_A$,

$$c_{k_i} = \langle c_{k_\alpha} \rangle^k + s_i \langle j_{k_i} \rangle_A. \quad [39]$$

Defining a surface-averaged concentration for Group α ,

$$\langle c_{k_\alpha} \rangle_A \equiv \frac{1}{A_\alpha} \sum_{i \in G_\alpha} \int_{A_i} c_{k_i} dA, \quad [40]$$

and inserting Eq. 39 gives,

$$\langle c_{k_\alpha} \rangle_A = \langle c_{k_\alpha} \rangle^k + \frac{1}{A_\alpha} \sum_{i \in G_\alpha} \langle j_{k_i} \rangle_A \int_{A_i} s_i dA. \quad [41]$$

We use this to define a new estimation of reaction rate, $j_{k_\alpha}^{\text{mc}} = f(\langle c_e \rangle^e, \langle \phi_e \rangle^e, \langle \phi_k \rangle^k, \langle c_{k_\alpha} \rangle_A)$, which is defined using the standard Butler-Volmer expression (as in Eq. 15), but with $\langle c_{k_\alpha} \rangle_A$ as the input for surface concentration. Analogously to Part I, we employ the assumption $\langle j_{k_i} \rangle_A \approx j_{k_\alpha}^{\text{mc}}$,

$$\langle c_{k_\alpha} \rangle_A = \langle c_{k_\alpha} \rangle^k + j_{k_\alpha}^{\text{mc}} \frac{1}{A_\alpha} \sum_{i \in G_\alpha} \int_{A_i} s_i dA. \quad [42]$$

Now that a correction on the surface concentration has been built, our final macroscale material balance can be written for Group α , by inserting $\langle j_{k_i} \rangle_A \approx j_{k_\alpha}^{\text{mc}}$ into Eq. 36,

$$\partial_t \langle c_{k_\alpha} \rangle^k = - \frac{A_\alpha}{FV_{k_\alpha}} j_{k_\alpha}^{\text{mc}}. \quad [43]$$

Validation methodology.—To demonstrate the MC model (herein abbreviated to just MC), we parameterise and solve it for Electrode C from Part I; the electrode and its closure field are shown in Fig. 7. Alongside MC, DFN, DC^y, DC^{1t} and DC^{1r} are included for comparison. All methodology is identical to that presented in Part I (material parameters, cycling profile, solver choices) and thus is not repeated here. To solve MC, PyBaMM's³³ custom modelling functionality was used, with an identical discretisation and solver to the DC models. To define the particle groups, we first order the particles by volume. Particles are then assigned to groups with the objective of obtaining a similar number of particles in each group. Alternatively, particles can be assigned so as to have similar values of V_{k_α} or A_α across the groups, but this was not found to have a significant effect on results. We choose to define 5 particle groups, thus giving 5 material balance PDEs for AM concentration.

Results and discussion.—The cell-voltage results for all models are shown in Fig. 8. The accuracy of MC is considerably greater than that of all DC models. It is able to almost perfectly trace the DNS results, with the exception of a small error occurring in the relaxation regime. This improvement is due to MC relaxing two assumptions which are necessary in the DC models. Firstly, in the development of the latter, differences between average AM concentration across a *particle* and average AM concentration *globally* are considered small—this is manifested in $(\langle c_{k_i} \rangle^k - \langle c_k \rangle^k) \ll s_i \langle j_{k_i} \rangle_A$. In MC on the other hand, it is only the differences between the particle average and the *group* average which are considered small, $(\langle c_{k_i} \rangle^k - \langle c_{k_\alpha} \rangle^k) \ll s_i \langle j_{k_i} \rangle_A$. The second improvement of MC is analogous to this, but on the question of reaction rate. Rather than assuming that the particle surface average of reaction rate is similar to the global surface average, $\langle j_{k_i} \rangle_A \approx \langle j_k \rangle_A$, it is instead considered similar to the group surface average $\langle j_{k_i} \rangle_A \approx \langle j_{k_\alpha} \rangle_A$. Electrode C is ideal for demonstrating this advantage, as a range of particle radii are present. The MC model is able to capture the reality that average AM concentration and average reaction rate may be significantly different between particles of different sizes. Furthermore, the MC model remained lighter to solve compared to the DFN—for the simulation here, MC took 0.177 s, compared to 0.283 s for the DFN.

Some analogy could be drawn between MC and the multi-particle DFN model (MP-DFN) presented by Kirk et al.³⁴ However, MC maintains the advantages that it makes no assumption on particle shape, and can be calculated directly from the AM microstructure. As a final aside, introducing a transient closure to MC is not as

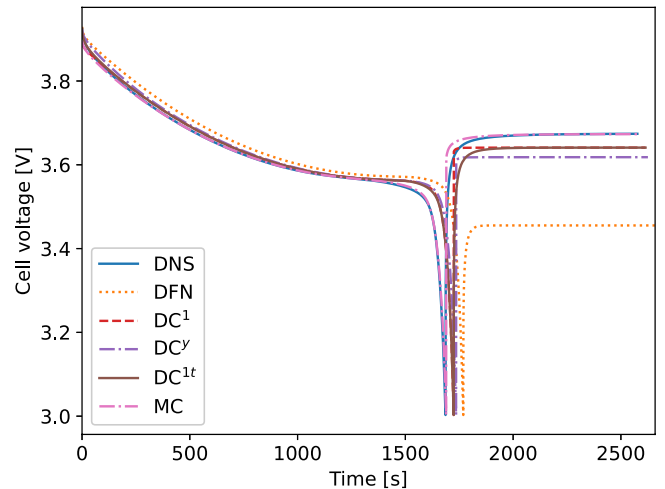


Figure 8. Cell-voltage against time a discharge-relaxation simulation on Electrode C from Part I, showing the DNS, DFN, DC^y, DC^{1t}, DC^{1r} and MC.

straightforward as DC^1 , since approximating $\langle j_{k\alpha} \rangle_A$ as a Heaviside step function is not reasonable, unlike for $\langle j_k \rangle_A$.

Single-Particle and Single-Continuum Models

Theory.—In this section, we present a “single-particle version” of both DC^1 and DC^{1t} . The key idea undergirding the SPM is that the different particles within an electrode behave similarly, and can thus be represented by a single representative particle. The DC model can be reduced in the same spirit, by assuming that $\langle c_k \rangle^k$ does not vary significantly along the thickness of the electrode, x . Rather than producing a single representative particle for each electrode (in which a PDE must be solved), this results in a single ordinary differential equation to describe the AM in each electrode, since $\langle c_k \rangle^k(x, t)$ becomes $\langle c_k \rangle^k(t)$. The models are therefore dubbed “single continuum” (SC), to indicate that there is now only a single continuum PDE defined across the thickness of the electrode (that for $\langle c_e \rangle^e$). The reduced versions of DC^1 and DC^{1t} are SC^1 and SC^{1t} . We emphasise that the way in which AM concentration is calculated is the *only* difference between the formulations of SC^1 , SC^{1t} , and the SPM.

We will compare our models to a version of the SPM published by Marquis et al.¹² known as the SPMe, where “e” symbolises that the model includes electrolyte dynamics—this is also the default SPMe model used in PyBaMM.³³ We begin by presenting the equations shared by SPMe, SC^1 , and SC^{1t} . All symbols used in this section are as defined in Part I of this work. A key assumption made in SPMs is that the intercalation reaction is uniformly distributed across the electrode surface,

$$j_k = \lambda_k \frac{i_{app}}{L_k a_k}, \quad [44]$$

where $\lambda_n = -1$ and $\lambda_p = 1$. Note that we employ the convention used in solvers such as BEST,^{35,36} which is the opposite of that used in Brosa-Planella’s work,¹⁵ in which λ_n and λ_p are reversed. Also, note that a superscript is no longer used on j_k (e.g. j_k^1) since its value is identical for each model. The form of the electrolyte transport equation is unchanged from that in the DFN and DC models,

$$\varepsilon_e \partial_t \langle c_e \rangle^e = \frac{\partial}{\partial x} \left(D_e \mathfrak{B}_e \frac{\partial \langle c_e \rangle^e}{\partial x} \right) + F^{-1} (1 - t_+^0) a_k j_k, \quad [45]$$

for $0 \leq x \leq L$, where

$$\varepsilon_e = \begin{cases} \varepsilon_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \varepsilon_{e,p} & \text{for } L_s < x \leq L, \end{cases} \quad [46]$$

$$\mathfrak{B}_e = \begin{cases} \mathfrak{B}_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \mathfrak{B}_{e,p} & \text{for } L_s < x \leq L, \end{cases}$$

$$a_k = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ a_p & \text{for } L_s < x \leq L, \end{cases} \quad [47]$$

$$j_k = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ j_p & \text{for } L_s < x \leq L. \end{cases}$$

However, we can see that Eq. 44 leads to the decoupling of the microscale and homogenised PDEs present in the DFN model, since the source term in Eq. 45 ($a_k j_k$) is no longer a function of AM concentration. As for the previous simulations, we consider a half cell, and thus the negative electrode is not modelled by this equation. The BCs and initial condition are,

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = -\frac{1}{F \mathfrak{B}_{e,s} D_e} (1 - t_+^0) i_{app} \quad \text{at } x = 0, \quad [48a]$$

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = 0 \quad \text{at } x = L, \quad [48b]$$

$$\langle c_e \rangle^e = c_{e,0} \quad \text{at } t = 0. \quad [48c]$$

In all models, Eq. 45 is solved numerically. Since $\langle c_e \rangle^e$ and j_k are now known, the equation for $\langle \phi_e \rangle^e$ can then be integrated and solved *a posteriori* (see the Appendix). The cell voltage for each model is calculated using a sum of overpotential contributions, as summarised by Brosa-Planella,¹⁵

$$V(t) = U_k - \eta_r - \eta_c - \Delta \phi_e - \Delta \phi_s, \quad [49]$$

where

$$\eta_r = \frac{2RT}{F} \left(\operatorname{arcsinh} \left(\frac{j_n}{2j_{n0,av}} \right) - \operatorname{arcsinh} \left(\frac{j_p}{2j_{p0,av}} \right) \right), \quad [50a]$$

$$\eta_c = \frac{2RT(1 - t_+^0)}{F} (1 + \delta_e) \left(\frac{1}{L_n} \int_0^{L_n} \ln \left(\frac{\langle c_e \rangle^e}{c_{e,av}} \right) dx - \frac{1}{L_p} \int_{L-L_p}^{L_p} \ln \left(\frac{\langle c_e \rangle^e}{c_{e,av}} \right) dx \right), \quad [50b]$$

$$\Delta \phi_e = -\frac{i_{app}}{\sigma_e} \left(\frac{L_n}{3\mathfrak{B}_{e,n}} + \frac{L_s}{\mathfrak{B}_{e,s}} + \frac{L_p}{3\mathfrak{B}_{e,p}} \right), \quad [50c]$$

$$\Delta \phi_s = -\frac{i_{app}}{3\sigma_s} \left(\frac{L_p}{\mathfrak{B}_{s,p}} + \frac{L_n}{\mathfrak{B}_{s,n}} \right), \quad [50d]$$

and

$$j_{n0,av} = \frac{1}{L_n} \int_0^{L_n} j_{n0} dx, \quad [51a]$$

$$j_{p0,av} = \frac{1}{L_p} \int_{L-L_p}^L j_{p0} dx, \quad [51b]$$

$$c_{e,av} = \frac{1}{L} \int_0^L \langle c_e \rangle^e dx. \quad [51c]$$

We have written the overpotential contributions in a general form, but these can be simplified in our case since we are studying a half cell (see the Appendix). The differences between each model arise in the calculation of AM surface concentration, $\langle c_k \rangle_A^m$, where m identifies the model, $m \in \{\text{spm}, \text{sc}^1, \text{sc}^{1t}\}$, denoting the SPMe, SC^1 , and SC^{1t} respectively. This appears in the open circuit potential, $U_k(\langle c_k \rangle_A^m)$, and also in the exchange current density,

$$j_{k0} = F k_k \sqrt{\langle c_k \rangle_A^m \langle c_e \rangle^e (c_{k,max} - \langle c_k \rangle_A^m)}. \quad [52]$$

For SPMe, $\langle c_k \rangle_A^{\text{spm}} = c_k|_{r=R_k}$, which comes from the solution of the PDE in the representative sphere,

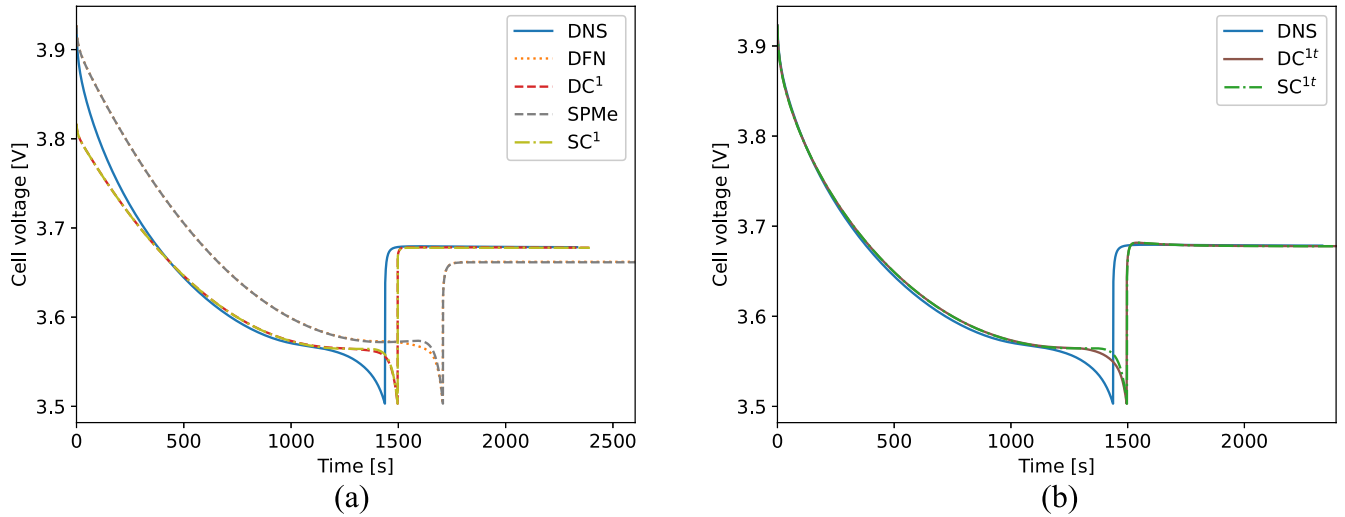


Figure 9. The simulation shown in Fig. 3b (and Fig. 6b in Part I), repeated for SC^1 and SPMe (a), followed by SC^{1t} (b).

$$\partial_t c_k = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_k \frac{\partial c_k}{\partial r} \right) \quad \text{for } 0 \leq r \leq R_k, \quad [53a]$$

$$-D_k \frac{\partial c_k}{\partial r} = \frac{j_k}{F} \quad \text{at } r = R_k, \quad [53b]$$

$$-D_k \frac{\partial c_k}{\partial r} = 0 \quad \text{at } r = 0, \quad [53c]$$

whilst for SC^1 and SC^{1t} we have

$$\langle c_k \rangle_A^{sc1} = \langle c_k \rangle^k + \langle s \rangle_A j_k, \quad [54a]$$

$$\langle c_k \rangle_A^{sc1t} = \langle c_k \rangle^k + \langle s^t \rangle_A \star \partial_t j_k, \quad [54b]$$

and

$$\partial_t \langle c_k \rangle^k = -\frac{A}{F V_k} j_k. \quad [55]$$

Results.—To compare the models, we use the simulation case introduced in Fig. 3b (and in Fig. 6b of Part I). This scenario employs the fractured-particle electrode from Part I (Electrode B) together with a low AM diffusivity, which amplifies the differences between models (see Part I for further discussion).

Figure 9a shows the results for the SPMe and SC^1 models, with DNS, DFN, and DC^1 included for reference. The predictions of SC^1 and DC^1 are almost indistinguishable, while SPMe also reproduces DFN very closely. Hence, neglecting variations in AM concentration across the electrode thickness—an assumption shared by both SPMe and SC^1 —has little impact on the predicted voltage in this case. Toward the end of discharge, the non-reduced models (DFN and DC^1) offer a slight improvement, since the estimation of surface concentration is more influential in this region.

The same conclusions hold for the models with transient closure (Fig. 9b): SC^{1t} captures DC^{1t} very accurately. Moreover, when compared with the DNS reference, SC^{1t} is substantially more accurate than DC^1 and DFN, particularly at the start of the discharge.

Our justification of SC^1 's and SC^{1t} 's improved accuracy relative to the SPMe follows the same reasoning as our comparison with the DFN in Part I—we abandon the assumption of spherical particles in the SC models, and include the transport characteristics of the real microstructure via the closure problem.

Table II. The computation times for the simulations shown in Fig. 9.

Model	Computation time [s]
DFN	0.589
DC^1	0.179
DC^{1t}	0.160
SPMe	0.0420
SC^1	0.0386
SC^{1t}	0.0406

Computation times for all models are summarised in Table II (numerical details as in Part I SM). SC^1 and SC^{1t} require approximately 25% of the runtime of their non-reduced counterparts (DC^1 and DC^{1t}). Notably, SC^{1t} is particularly interesting when compared with the DFN: it offers improved accuracy while requiring only about 7% of the DFN computation time for this case.

Conclusions and Perspectives

In the second instalment of this work, four extensions to the dual-continuum (DC) model developed in Part I have been presented. Collectively, these extensions follow a progression of increasing complexity, broadening generality, and improving efficiency. Validations were performed by comparison with a detailed microscale model, the ubiquitous Doyle–Fuller–Newman (DFN) model,³⁷ and another DC model recently published by Yang and Tartakovsky.⁷ Firstly, we revisited the closure problem, which provides a mapping between volume-averaged and surface-averaged active material (AM) concentration. Rather than using the “quasi-steady-state” assumption (as in Part I), we considered the transient behaviour of the closure variable. This improved accuracy under changing battery loads (at the beginning of discharge and during the relaxation regime). Secondly, we extended the microscale model of Part I to include a carbon-binder domain (CBD), and tested four electrodes built from a tomographic image, with varying CBD volume fraction and arrangement. The inclusion of a boundary between AM and CBD within the closure problem enabled our DC model to precisely simulate microstructures with CBD. Thirdly, we generalised the DC theory to a multi-continuum (MC) model, in which several macroscale equations are used to describe the AM phase. The MC model was validated for an electrode with a large particle size distribution, and enhanced accuracy notably towards the end of the discharge and during the relaxation period. Finally, we reduced the DC theory to obtain two single-continuum (SC) models—one using the steady-state closure and the other retaining transient behaviour. Both

models preserved the accuracy of the original DC formulation while providing a further reduction in computational cost.

A particularly interesting perspective of this work is the reconstruction of microscale fields. Using the closure variable, which is defined over the entire electrode microstructure, concentration fields within the real AM geometry can be estimated. This enables the prediction of local concentration fluctuations without the need to solve a detailed microscale model, offering a computationally efficient tool that may prove especially useful in ageing studies.

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Appendix

A.1. The Dimensionless Formulation of the Transient Closure Problem.—The following dimensionless groups are defined,

$$s_i^t = \frac{L}{D_k F} \hat{s}_i^t, \quad [56a]$$

$$t = \hat{t} \frac{L^2}{D_k}, \quad [56b]$$

$$x = \hat{x}L, \quad [56c]$$

where L is the length of the electrode. Using these in Eq. 8 (with the inclusion of the BC at $\partial\Omega_{k,\text{cbd}}$) gives,

$$\partial_{\hat{t}} \hat{s}_i^t = \hat{\Delta} \hat{s}_i^t + \frac{L A_i}{V_{k_i}} \quad \text{in } \Omega_{k_i}, \quad [57a]$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i^t = -1 \quad \text{at } \partial\Omega_{k_i,e}, \quad [57b]$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i^t = -\varepsilon_{e,\text{cbd}}^{\text{surf}} \quad \text{at } \partial\Omega_{k_i,\text{cbd}}, \quad [57c]$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i^t = \mathbf{n} \cdot \hat{\nabla} \hat{s}_j^t \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [57d]$$

$$\hat{s}_i^t = \hat{s}_j^t \quad \text{at } \partial\Omega_{k_i,k_j}, \quad [57e]$$

$$\hat{s}_i^t = 0 \quad \text{at } t = 0. \quad [57f]$$

Additional Information for the SPM-Type Models.—In the SPM_e, SC¹, and SC^{1t} models, $\langle \phi_e \rangle^e$ is governed by

$$\frac{\partial \bar{i}_e}{\partial x} = a_k j_k \quad \text{for } 0 < x < L, \quad [58]$$

where

$$\begin{aligned} \bar{i}_e &= -\sigma_e \mathfrak{B}_e \frac{\partial \langle \phi_e \rangle^e}{\partial x} + \sigma_e \mathfrak{B}_e \frac{2RT(1 - t_+^0)}{F} \\ &\times (1 + \delta_e) \frac{\partial \ln \langle c_e \rangle^e}{\partial x}, \end{aligned} \quad [59]$$

with the half-cell BCs

$$\bar{i}_e = i_{\text{app}} \quad \text{at } x = 0, \quad [60]$$

$$\bar{i}_e = 0 \quad \text{at } x = L. \quad [61]$$

Any symbols not defined above are as defined in the SM to Part I. The overpotential contributions given in Eq. 50 can be simplified for the case of a half cell,

$$\eta_r = \frac{2RT}{F} \left(\text{arcsinh} \left(-\frac{i_{\text{app}}}{2j_{li,0}} \right) - \text{arcsinh} \left(\frac{j_p}{2j_{p0,\text{av}}} \right) \right), \quad [62a]$$

$$\begin{aligned} \eta_c &= \frac{2RT(1 - t_+^0)}{F} (1 + \delta_e) \left(\ln \left(\frac{\langle c_e \rangle^e|_{x=0}}{c_{e,\text{av}}} \right) \right. \\ &\quad \left. - \frac{1}{L_p} \int_{L-L_p}^{L_p} \ln \left(\frac{\langle c_e \rangle^e}{c_{e,\text{av}}} \right) dx \right), \end{aligned} \quad [62b]$$

$$\Delta \phi_e = -\frac{i_{\text{app}}}{\sigma_e} \left(\frac{L_s}{\mathfrak{B}_{e,s}} + \frac{L_p}{3\mathfrak{B}_{e,p}} \right), \quad [62c]$$

$$\Delta \phi_s = -\frac{i_{\text{app}}}{3\sigma_s} \frac{L_p}{\mathfrak{B}_{s,p}}, \quad [62d]$$

where $j_{li,0}$ is the exchange current density for the lithium foil.

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